

Lab-4

Practical Exercise: Build and Run a Java Application with Maven, Migrate the Same Application to Gradle

Step 1: Creating a Maven Project

- Already we have created maven project in experiment 2. Go to experiment 2 pom.xml folder. In the folder pom.xml file, src folder and .mvn and target folder should be there.
- Copy the path of pom.xml. Open the command prompt. Change directory to pom.xml folder.

Step 2: Migrate the Maven Project to Gradle

1. **Initialize Gradle:** Navigate to the project directory (**gradle-example**) and run:
gradle init
 - It will ask **Found a Maven build. Generate a Gradle build from this? (default: yes) [yes, no]**
 - Type **Yes**
 - **Select build script DSL:**
 - 1: Kotlin
 - 2: Groovy
 - Enter selection (default: Kotlin) [1..2]
 - Type **2**
 - **Generate build using new APIs and behavior (some features may change in the next minor release)? (default: no) [yes, no]**
 - Type **No**
2. Navigate the project folder and open **build.gradle** file then add the below code and save it.

```
plugins {  
  
    id 'java-library'  
  
    id 'maven-publish'  
  
}  
  
repositories {
```

```

mavenLocal()

maven {

    url = uri('https://repo.maven.apache.org/maven2/')

}

}

dependencies {

    testImplementation 'junit:junit:3.8.1'

}

group = 'com.yit'

version = '1.0-SNAPSHOT'

description = 'maven-demo'

java.sourceCompatibility = JavaVersion.VERSION_1_8

publishing {

    publications {

        maven(MavenPublication) {

            from(components.java)

        }

    }

}

task run(type: JavaExec) {

    main = 'com.yit.App'

    classpath = sourceSets.main.runtimeClasspath }

```

Step 3: Run the Gradle Project

- **Build the Project:** In the project directory (gradle-example), run the below command to build the project:

gradlew build

- **Run the Application:** Once the build is successful, run the application using below command:

gradlew run

Step 4: Verify the Migration

- **Compare the Output:** Make sure that both the **Maven** and **Gradle** builds produce the same output:

- **Gradle Output:**

- **Maven Output:**

mvn clean install

mvn exec:java -Dexec.mainClass="com.yit.demo.App"